

Table 4: Descriptive Statistics of Alternative Methodologies for GENOE

	Baseline GENOE	Blind to occup. name	Task-by-task	Ethical emphasis	Ethical exclusion	LLMs	Non-AI technologies	Only AI
mean	0.382	0.391	0.560	0.370	0.423	0.204	0.343	0.039
p25	0.245	0.283	0.446	0.238	0.277	0.100	0.216	0.005
p50	0.317	0.325	0.546	0.312	0.355	0.175	0.295	0.031
p75	0.475	0.469	0.680	0.438	0.594	0.261	0.399	0.068
min	0.095	0.117	0.250	0.098	0.079	0.009	0.089	-0.108
max	0.932	0.862	0.891	0.885	0.945	0.857	0.919	0.391
Corr with baseline	1.000	0.947	0.888	0.992	0.982	0.475	0.946	0.346
Corr with FN	-0.081	-0.043	0.065	-0.102	-0.082	0.661	-0.261	0.507
Corr with PZ	0.569	0.559	0.616	0.560	0.564	0.666	0.430	0.511
Corr with WB	0.072	0.061	0.066	0.061	0.062	-0.036	0.048	0.085
Corr with FO	0.545	0.501	0.470	0.546	0.553	0.029	0.613	-0.082
P-value of diff with baseline	.	0.000	0.000	0.000	0.000	0.000	0.000	0.000

Source: Authors’ calculations using data from O*NET and surveys conducted with GPT-4o through OpenAI’s API. *Note:* This table shows the descriptive statistics of the alternative indexes at the SOC-2010 occupational level for the five-year horizon, the correlation with the baseline GENOE and other existing indexes, and the p-value of the mean difference with the baseline GENOE. FN refers to Felten *et al.* (2021), PZ to Pizzinelli *et al.* (2023), WB to Webb (2019) and FO to Frey and Osborne (2017).

These alternative methodologies serve a dual purpose: they validate the robustness of our baseline GENOE index while offering deeper insights into the complex landscape of AI’s potential impact on the workforce. By examining occupational exposure through various lenses—from task-level granularity to ethical considerations and AI-specific advancements—we gain a more comprehensive understanding of the challenges and opportunities that lie ahead. Our findings underscore the necessity of a multifaceted approach in assessing the future of work, one that considers not only the technical capabilities of AI but also the contextual, ethical, and broader technological factors that shape occupational vulnerability. This perspective is crucial for policy-makers, employers, and workers as they navigate the evolving dynamics of human-AI interaction in the workplace and prepare for a future where the boundaries between human and artificial intelligence continue to shift.

3.4 Replication of Felten *et al.* (2021)’s AIOE Using Synthetic Surveys

To validate the effectiveness of synthetic surveys in producing assessments comparable to those of human experts, we replicated the well-established AI Occupational Exposure Index (AIOE) developed by Felten *et al.* (2021), which has been widely used in various empirical applications

(Acemoglu *et al.*, 2022; Fossen and Sorgner, 2021).⁷ In our replication, we replaced the original human expert surveys with synthetic surveys conducted using GPT-4o via OpenAI’s API.

The AIOE measures the degree of overlap between advancements in AI and workplace abilities. They define 10 key AI applications from the Electronic Frontier Foundation, which have seen considerable development since 2010: abstract strategy games, real-time video games, image recognition, visual question answering, image generation, reading comprehension, language modeling, translation, and speech recognition. They then link these AI applications to the 52 labor abilities recorded in O*NET by conducting 2,000 surveys with gig workers from Amazon’s Mechanical Turk (MTurk).⁸ Each participant was asked whether they believe each application is related to each ability, with a yes or no answer. These responses were processed as binary variables and averaged across all respondents to construct a matrix of relatedness, with a score between 0 and 1 for each application and each ability. The ability-level exposure was calculated as the sum of the scores of all applications for Ability j (A_j). Finally, they calculated the occupation-level ($AIOE_k$) for every occupation k as follows:

$$AIOE_k = \frac{\sum_{j=1}^{52} A_j \times L_{jk} \times I_{jk}}{\sum_{j=1}^{52} L_{jk} \times I_{jk}} \quad (1)$$

Using the replication files for the AIOE made available online by the authors, we re-estimated the $AIOE_k$ by replacing the matrix built with mTurk surveys with a vector resulting from surveys conducted with GPT-4o through OpenAI’s API. Instead of defining the 10 AI applications, we asked the language model the following: “Based on recent advancements in AI, please answer ‘Yes’ or ‘No’ depending on whether you believe that AI is related to or could be used for the following ability: [ability name] [ability description].” This was done for the 52 abilities defined in O*NET and repeated 100 times, with the temperature setting randomized. Although this approach differs from the original methodology by not specifying 10 applications, we made this change to minimize conditioning the model. This method likely results in a more conservative estimate of the index by reducing potential similarities between the original and replicated indexes.

Since we do not have specific AI applications, we calculated the A_j as 10 times the average score from the 100 iterations. This adjustment was made solely for comparability purposes and does not affect the index calculation. We then ran the same equation (1) and computed the new index.

The results from the estimation, and the comparison with the original index, can be observed in Figure 3. The correlation between both indexes is 0.99, indicating a very high degree of replicability and very small variations between the answers from the mTurk workers and the answers

⁷The AIOE was initially introduced in Felten *et al.* (2018) and subsequently modified and updated in Felten *et al.* (2021).

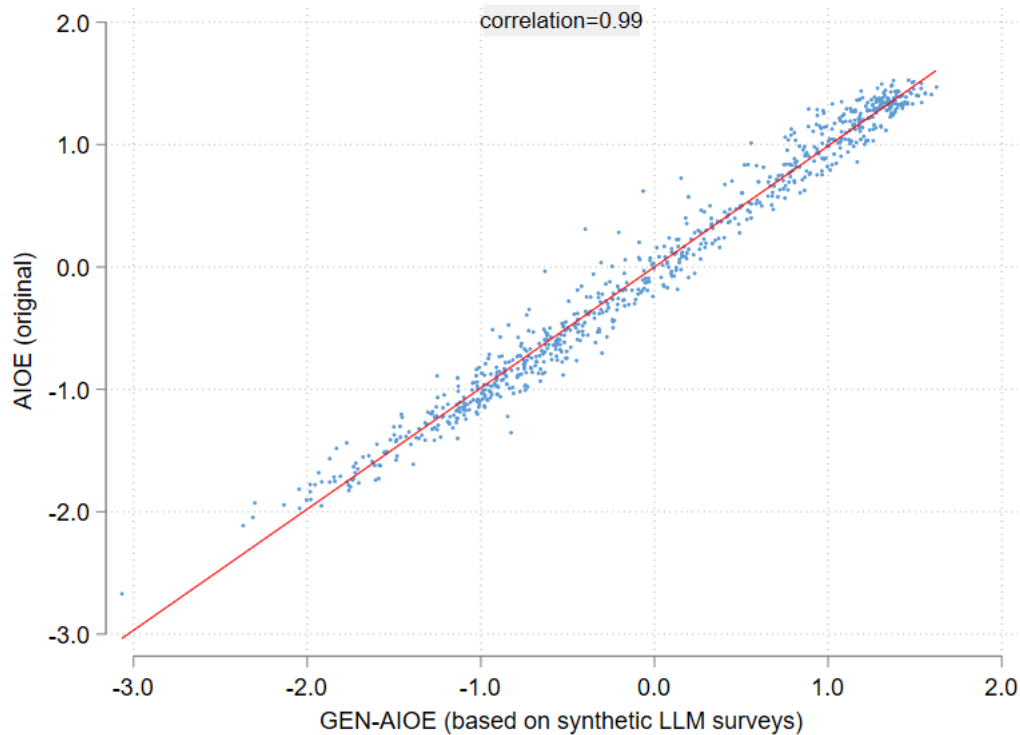
⁸Amazon Mechanical Turk (MTurk) is an online crowdsourcing platform that allows businesses to outsource a wide range of tasks to a global, distributed workforce. These tasks can vary from data validation and research to content moderation and survey participation.

provided by the synthetic surveys with GPT-4o.

A potential concern might be that GPT-4o could generate similar responses to mTurk workers because it searches the web or because its training data includes information from the AIOE and the mTurk surveys. However, GPT-4o does not perform real-time web searches unless explicitly instructed to do so, and in our study, we did not prompt GPT-4o to search the web. Regarding the second concern of the training data, as shown in follow-up surveys from the GENOE, the reasoning of the model is more holistic, based on its understanding of AI advancements and specific labor and workplace tasks (in this case, abilities), and does not rely on specific unique sources. Therefore, while it is possible that the training data could include some pre-existing information on Felten *et al.* (2021)'s work, among millions of other sources, the responses are synthesized from a broader knowledge base and not directly copied from any particular source.

Therefore, the high correlation observed suggests that the model's responses are consistent with human judgment for this specific task. This consistency supports the robustness and reliability of using artificial intelligence tools to emulate human analytical capabilities in these types of assessments.

Figure 3: Correlation of Original AIOE and AIOE Built with Synthetic Surveys



Source: Authors' formulation based on data from O*NET and Felten *et al.* (2021) *Note:* Each dot represents an occupation at the SOC 10-digit code.

4 OCCUPATIONAL EXPOSURE IN THE U.S. AND MEXICO LABOR MARKETS

A key strength of the GENOE index is its capacity for a detailed analysis of occupational exposure to AI-driven automation across different labor markets. In this section, we use the index to examine two distinct contexts: the United States, the world's largest developed economy, and Mexico, one of the largest emerging market economies.

We retrieved data from the U.S. Bureau of Labor Statistics' Current Population Survey (CPS) and Mexico's INEGI Encuesta Nacional de Ocupación y Empleo (ENOE), focusing on surveys from January to December 2023. We mapped the GENOE estimates from the SOC 10 classification to each survey using the respective employment classifications provided by each survey, with crosswalks from the U.S. Bureau of Labor Statistics and the International Labour Organization (ILO). Our analysis concentrated on workers between 15 and 65 years old.

Figure 4 displays the distribution of jobs in the United States and Mexico according to the GENOE index across different time horizons. For the one-year horizon, 81% of jobs in the United States fall into the lower exposure categories (0 to 0.4), compared to 83% in Mexico. In the middle exposure category (0.4 to 0.6), Mexico has a slightly higher concentration of jobs (13%) compared to the United States (10%). In the high exposure categories (0.6 to 1), there is a low concentration of jobs in both countries, with 9% in the United States and only 5% in Mexico, and virtually no jobs at the highest risk level (0.8 to 1) in both countries.

As the time horizon extends, a shift towards higher exposure becomes evident for both countries. Looking at the five-year horizon, the share of jobs in the low exposure categories decreases to 59% in the United States and 54% in Mexico. The middle exposure category increases to 21% in the United States and 29% in Mexico. The high exposure categories remain relatively low, with 20% in the United States and 17% in Mexico, indicating a strong increase in vulnerability over this time frame.

The long-term outlook shows a significant increase in potential AI impact. In the 10-year horizon, the low exposure categories comprise 51% of jobs in the United States and 43% in Mexico, while the high exposure categories increase to 33% and 30% in respectively. This showcases a significant increase in potential AI impact over the long term, though the highest risk level (0.8 to 1) remains low at 6% in the United States and 1% in Mexico.